Pavan Gorintla

PhD in Informatics | Playful Health Technology Lab | Human-Computer Interaction | Accessibility | Serious Games for Health | MS Graduate from Northern Arizona University | PGDMA Graduate from Alliance University

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Research Summary

I am a PhD researcher in Informatics at Northern Arizona University, working in the Playful Health Technology Lab under Dr. Jared Duval. My research sits at the intersection of human-computer interaction, accessibility, data visualization, and playful health technology. I build interactive, accessible research tools and community co-designed health interventions for populations historically underserved by digital health systems. My current dissertation work spans CardioCare Quest, an NIH-funded community playful telehealth intervention for hypertension management; ongoing research on an assistive robotic arm; NetGauge RITE sessions for rapid iterative evaluation of a crowdsourced broadband measurement platform; an interactive accessible visualization platform organizing three decades of accessibility and serious-games research; contribution to SpokeIt, a mobile speech-therapy game for children born with orofacial cleft; and a machine-learning literature review pipeline. Three years of prior industry software development ground my research in the practical constraints of shipping accessible, reliable tools.

Education

Doctor of Philosophy, Informatics and Computing *Jan 2025 – Present*

Northern Arizona University

Advisor: Dr. Jared Duval

Current research projects I am working on in the Playful Health Technology Lab:

- **CardioCare Quest.** Community playtesting of a playful telehealth intervention for hypertension management. Currently running a study with adults who have intellectual and developmental disabilities at the Hozhoni Foundation.
- **Assistive Robotic Arm.** Ongoing research on playful, accessible interactions with an assistive robotic arm for users with upper-limb mobility impairments. I focus on the human-computer interaction layer.
- **NetGauge RITE Sessions.** Rapid Iterative Testing and Evaluation sessions for NetGauge, a platform of co-created games that make measuring broadband quality enjoyable and contribute to crowdsourced coverage maps.
- **Interactive Data Visualization.** Lead developer of an accessible web platform that turns a systematic review of 215 accessibility and serious-games papers (1994-2023) into an explorable design-space map.
- **SpokeIt Player Registration.** Contributed the mobile onboarding and player registration screens for SpokeIt, an NIH-supported speech-therapy game for children born with orofacial cleft.
- **Machine-Learning Literature Review.** Co-author on a retrieval-augmented pipeline that uses four language models to annotate 215 papers against a human consensus baseline of 1,720 annotations.

Master of Science, Computer Science *Aug 2022 – Dec 2023*

Northern Arizona University

- Studied algorithmic analysis and the importance of choosing the right data structure for different scenarios.
- Explored transaction management, query optimization, and database security.
- Learned software development methodologies such as Agile and Waterfall, emphasizing systematic software design, development, testing, and maintenance.
- Explored software architecture patterns and design principles for scalable and maintainable software systems.
- Conducted in-depth research on a specific topic within computer science, applying theoretical knowledge to practical problems.
- Developed critical thinking, problem-solving, and academic writing skills through the completion of a master's thesis.

Post Graduate Diploma in Management, Information Systems and Services *Jan 2020 – Jan 2022*

Alliance University, Bengaluru, India

- Explored the integration of strategic management principles with information systems to align technology with organizational goals.
- Studied IT governance frameworks and their role in decision-making and risk management.
- Learned techniques for analyzing large datasets to extract meaningful insights for business decision-making, applying statistical methods, data visualization, and predictive modeling to real-world business scenarios.
- Explored the implementation and customization of ERP systems to streamline business processes across functional areas, including finance, supply chain management, and human resources.
- Acquired IT project management skills including planning, scheduling, budgeting, and risk management; applied Agile and Waterfall methodologies to manage projects effectively.
- Studied cybersecurity threats, vulnerabilities, and risk assessment techniques; learned security frameworks, cryptography, and compliance standards such as GDPR and ISO 27001.

Bachelor of Technology, Computer Science and Engineering *Jun 2015 – May 2019*

Sreenivasa Institute of Technology and Management Studies, Chittoor, India

- Learned core programming languages including C, C++, and Java, building a strong foundation in software development and algorithmic problem-solving.
- Studied the principles of computer networks, including TCP/IP, LAN, WAN, wireless networks, and communication technologies.
- Acquired skills in web development technologies (HTML, CSS, JavaScript) and server-side scripting (PHP); designed and developed dynamic web applications.
- Explored the fundamentals of operating systems, including process management, memory management, and file systems, with system programming in Unix/Linux environments.
- Studied software development lifecycle models, software requirements engineering, testing methodologies, and applied software quality assurance and debugging techniques in real-world projects.

Publications and Presentations

All three manuscripts below are in preparation for the ACM Transactions on Accessible Computing Special Issue on Play and Accessibility, submission deadline July 10, 2026.

First-Author Manuscripts in Preparation

- **Gorintla, P. P.**, Duval, J., and colleagues. "Community Playtesting CardioCare Quest Among Adults with Disabilities at the Hozhoni Foundation." In preparation for *ACM Transactions on Accessible Computing*, Special Issue on Play and Accessibility, 2026.
- **Gorintla, P. P.** and Duval, J. "An Interactive, Accessible Visualization Platform for Systematic Review of Playful Health Technology Research." In preparation for *ACM Transactions on Accessible Computing*, Special Issue on Play and Accessibility, 2026.

Co-Authored Manuscripts in Preparation

- Hagemann, S., Pokkula, S. K., **Gorintla, P. P.**, Beach, H., Somaskantha Iyer, S., Vigil-Hayes, M., Amresh, A., and Duval, J. "Applying Accountability and Transparency to an LLM-Powered Synthesis of Game-Based Research in ASSETS." In preparation for *ACM Transactions on Accessible Computing*, Special Issue on Play and Accessibility, 2026.

Academic Writing in Progress

- **Gorintla, P. P.** "Mapping Three Decades of Experiential Design in Playful Health Technology: A Systematic Review with Interactive Visualization." Comprehensive examination paper, Northern Arizona University.

Symposium Presentations

- **Gorintla, P. P.** and Duval, J. "What 30 Years of Accessibility-Focused Games Research Reveals About Health Game Design." Research Symposium Poster, Graduate Student Government, Northern Arizona University, 2025.

Honors and Awards

- **First Prize, NAU Engineering Fest 2024** for SpokeIt: Advancing Accessible and Engaging Speech Therapy. Faculty advisor: Dr. Jared Duval.
- Graduate Research Assistantship, Playful Health Technology Lab, Northern Arizona University. *2025 – Present*

Teaching Experience

Graduate Teaching Assistant and Grader *Aug 2023 – Present*

School of Informatics, Computing, and Cyber Systems, Northern Arizona University

Assisted Dr. Jared Duval in the Playful Health Technology Lab by grading assignments, providing feedback, and mentoring students across multiple computer science courses. Supported courses in the Immersive Media and Games program and the computer science department across Spring 2026, earlier semesters in Data Structures, and Principles of Database Systems. Spring 2026 assignments: Software Architecture and Open Source Software Development.

Responsibilities

- Reviewed and assessed student assignments and projects with a focus on ensuring adherence to academic standards.
- Provided constructive and personalized feedback aimed at enhancing students' understanding and encouraging

their academic growth.

- Held regular office hours to offer additional support, address individual student concerns, and assist with coding and problem-solving.
- Fostered an inclusive and interactive learning environment, encouraging students to participate in discussions and collaborative activities.
- Provided mentorship and guidance to help students navigate coursework and develop effective study strategies.

Impact

- Played a pivotal role in creating a positive and nurturing classroom atmosphere that contributed to improved student performance and engagement.
- Helped students build confidence in their abilities and develop a deeper interest in computer science.
- Gained insights into teaching methodologies and academic evaluation processes, enhancing my own pedagogical skills.
- Developed an understanding of diverse learning needs and tailored my approach to meet them effectively.

Professional Experience

UI/UX Developer *Mar 2024 – Dec 2024*

Playful Health Technology Lab, Northern Arizona University · Supervisor: Dr. Jared Duval

- Developed and maintained responsive web applications using HTML5, CSS3, and Bootstrap to ensure optimal viewing experiences across devices and screen sizes.
- Implemented media queries and flexible grid layouts to create mobile-first designs that enhanced user engagement and satisfaction.
- Created dynamic and interactive user interfaces using JavaScript and jQuery, improving user interaction and experience.
- Utilized AJAX to asynchronously fetch data from the server and update pages in real time, resulting in smoother, more responsive applications.
- Ensured cross-browser compatibility across Chrome, Firefox, Safari, and Edge by implementing and testing consistent behavior and addressing browser-specific issues.
- Collaborated with UI/UX designers to integrate visual and interactive elements, enhancing both aesthetic and functional quality.
- Conducted usability testing and incorporated user feedback to refine interface designs, leading to a more intuitive user experience.
- Optimized web application performance through code minification, image optimization, and efficient loading techniques, reducing page load times and improving retention.
- Implemented robust error handling to manage and display error messages gracefully, and conducted performance testing to ensure the platform ran efficiently.
- Used Git for version control and collaborated with team members through GitHub and GitLab, and conducted code reviews and regular team meetings to share knowledge and address challenges.

Java Developer *May 2019 – Aug 2022*

VIT Infotech, Bengaluru, Karnataka, India

Deeply involved in developing the Express Play project for the client Intertrust. Contributed to both front-end and back-end development, ensuring a seamless and user-friendly application.

Key Achievements

- **Front-End Development:** Crafted user-friendly interfaces with HTML, JSP, and Servlets, enhancing the overall user experience.
- **Client-Side Validation:** Used JavaScript to implement robust client-side validation, improving data accuracy.
- **Back-End Development:** Built Java applications with Spring Boot and Gradle, focusing on core Java and Spring Boot concepts.
- **Database Interaction:** Managed database operations with JPA and PostgreSQL and assisted in database migrations with Flyway.
- **API Documentation:** Used Swagger for clear API documentation.
- **CI/CD Systems:** Designed CI/CD pipelines with Jenkins and deployed using Docker and Kubernetes.
- **Microservices Development:** Developed RESTful microservices for seamless integration.
- **Project Management:** Followed SCRUM methodologies for efficient project management.
- **Client Collaboration:** Worked closely with Intertrust to deliver tailored solutions for Express Play, focusing on secure content distribution.

Web Developer (Internship) *Nov 2018 – Apr 2019*

R.K. Info Systems, Hyderabad, Telangana, India

Developed a web page for my college to facilitate interaction between students and professors about classwork and personal skill development.

Key Contributions

- Created a user-friendly layout that simplified navigation and encouraged active participation, making communication between students and professors more engaging and efficient.
- Implemented tools that facilitated discussions on class assignments and allowed students to share insights on skill-building activities, fostering a collaborative learning environment.
- Utilized responsive design techniques to guarantee optimal functionality and user experience across desktops, tablets, and smartphones, accommodating a diverse user base.
- Actively gathered and integrated feedback from users and stakeholders through usability testing, leading to iterative enhancements that improved site performance and user satisfaction.

Projects

CardioCare Quest: Community Playtesting Study *First author, 2025 – Present*

Lead researcher on a community playtesting study of CardioCare Quest, a playful telehealth intervention for hypertension management in Navajo, Black, and Hispanic communities, supported by NIH funding (5U54MD012388-08, U24MD015970) awarded to the Playful Health Technology Lab. The study is being conducted with adults who

have intellectual and developmental disabilities at the Hozhoni Foundation in Flagstaff, Arizona.

Responsibilities

- First author on the manuscript in preparation drawing on this study.
- Characterizing cardiovascular risk profiles, engagement patterns during open community play, and shifts in hypertension knowledge and self-monitoring attitudes after play.
- Coordinating the undergraduate capstone team building Twine-based mini-game prototypes.
- Working with a Figma designer on the clinician-facing dashboard.
- Contributing to protocol design, consent procedures, and accessibility accommodations for the community study.

Impact

- Generating research insights on how playful health technology can meaningfully engage adults with intellectual and developmental disabilities, a population under-served in digital health research.
- Contributing to the Playful Health Technology Lab's NIH-funded program on hypertension management in Indigenous and underserved communities.

Assistive Robotic Arm *Ongoing, 2025 – Present*

Ongoing research in the Playful Health Technology Lab on an assistive robotic arm, under Dr. Duval. The project investigates playful, accessible interactions for users with upper-limb mobility impairments. My contribution centers on the human-computer interaction layer, including interaction design and user-study methodology planning.

NetGauge: RITE Sessions for a Crowdsourced Broadband Platform *Contributor, 2025 – Present*

NetGauge is a lab platform of co-created games that make it enjoyable to measure broadband internet quality and contribute to crowdsourced broadband coverage maps, addressing a driver of rural and tribal health disparities. I contribute to Rapid Iterative Testing and Evaluation (RITE) sessions: participants use the platform until they encounter a usability problem, at which point the team iterates on the build before the next participant is run.

Responsibilities

- Facilitating RITE sessions and capturing usability problems in session notes.
- Supporting rapid iteration between sessions on the frontend of the platform.
- Contributing to accessibility review of the games in the NetGauge suite.

Interactive Accessible Visualization Platform for Research Synthesis *Lead developer, 2025 – Present*

Lead developer of a web platform that transforms a systematic review of 215 peer-reviewed papers on accessibility and serious-games research (1994-2023) into an interactive, accessible design-space explorer across nine experiential design dimensions.

Responsibilities

- Designed and implemented an interactive web interface using HTML, CSS, JavaScript, and D3.js to visualize complex data from a systematic review.

- Implemented coordinated spectrum heatmaps, dot matrices, and filtering views.
- Built full keyboard operability, screen-reader compatibility, and audio-tactile sonification from the start, meeting WCAG 2.1 AA contrast standards.
- Integrated dynamic features such as interactive charts, filters, and tooltips to allow users to explore data intuitively and gain insights.
- Collaborated with researchers to understand the key findings and data points that needed to be highlighted in the visualization.

Impact

- Conducted a pilot expert evaluation with eleven HCI researchers: nine of eleven preferred the heatmap view over bar charts, and participants completed exploratory synthesis tasks in roughly half the time of comparable manual review.
- Improved the presentation and accessibility of research data, making it easier for stakeholders, including those using assistive technologies, to interpret and utilize findings.
- Work anchors a first-author journal manuscript in preparation.

SpokeIt: Player Registration *Contributor, 2023 – 2024*

Contributed to SpokeIt, an NIH-supported mobile speech-therapy game for children born with orofacial cleft, under Dr. Duval. My specific contribution was on the player registration screens: the mobile onboarding flow through which a new player creates a profile before gameplay.

Responsibilities

- Implemented the player registration screens in the mobile app.
- Aligned the registration flow with the research team's consent and demographic-collection requirements.
- Tested the registration experience with the team and iterated on usability.
- Team member on the 2024 SpokeIt project that won First Prize at NAU Engineering Fest.

Machine-Learning Literature Review Pipeline *Co-author, 2025 – Present*

Co-author on a retrieval-augmented pipeline that runs four large language models against a corpus of 215 peer-reviewed papers on accessibility and serious games, comparing their answers to a human consensus baseline of 1,720 annotations from trained HCI researchers.

Responsibilities

- Contributed to prompt design, pipeline implementation, and statistical evaluation of model-human alignment (weighted quadratic Cohen's kappa, Kruskal-Wallis tests, Dunn's post-hoc analysis).
- Co-designed the decision-mode architecture so each model self-reports confidence, cites supporting source chunks, and iteratively expands retrieval through four modes (Full Context, Retrieval, Expanded Retry, Forced Choice) in service of fairness, accountability, and transparency.
- Manuscript in preparation as noted under *Publications*.

Express Play Project (Intertrust) *VIT Infotech, 2019 – 2022*

- Designed and implemented user-friendly interfaces using HTML, CSS, and Bootstrap, optimizing user experience for accessing DRM-protected content.
- Engineered scalable backend systems with Java, Spring Boot, and Gradle to support secure DRM content distribution.
- Implemented robust security measures, including Spring Security and encryption techniques, to protect DRM-secured content and user data.
- Managed database operations using JPA and PostgreSQL, ensuring efficient data storage and retrieval for DRM licensing and content distribution.
- Collaborated with cross-functional teams and applied Agile methodologies to deliver DRM licensing features and functionalities on schedule and within scope.

Web Page Development *R.K. Info Systems, 2018 – 2019*

- Gathered requirements from students and professors to understand their communication needs.
- Created wireframes to outline the platform's layout and features.
- Built a responsive platform using HTML, CSS, and Bootstrap for optimal viewing on all devices, with media queries to adapt the layout across screen sizes.
- Added interactive elements like real-time messaging and notifications using JavaScript, and utilized AJAX for smooth data loading without page refreshes.
- Worked with backend developers to connect the platform to APIs for data exchange, ensuring secure access with proper authentication methods.
- Collected feedback from users and implemented new features and enhancements based on their suggestions.

Technical Skills

- **Research computing:** Python, JavaScript, HTML, CSS, LaTeX, Overleaf, Git, GitHub.
- **Data visualization:** D3.js, Highcharts, Plotly; coordinated-views design; sonification for audio-tactile chart representation.
- **Applied large language models:** GPT, Gemini, Llama, Mistral APIs; retrieval-augmented generation with embedding models; prompt engineering (chain-of-thought, few-shot, structured output); inter-rater reliability evaluation of model-human alignment.
- **Accessibility practice:** WCAG 2.1 AA, ARIA patterns, keyboard-first interaction, screen-reader compatibility, sonification, audio-tactile charts.
- **Research methods:** PRISMA-based systematic review, qualitative thematic analysis, inductive coding, participatory and co-design methods, RITE sessions, expert evaluation, usability testing.
- **Statistical methods:** Weighted quadratic Cohen's kappa, Kruskal-Wallis tests, Dunn's post-hoc analysis, rank-biserial comparisons.
- **Design and prototyping:** Figma, Unity, Twine, Flutter, Firebase.
- **Programming languages:** Java, C, HTML, JSP, JavaScript, Python.
- **Frameworks:** Spring Boot, Gradle.

- **Tools and technologies:** Docker, Kubernetes, Jenkins, Swagger, Flyway.
- **Databases:** PostgreSQL, JPA.
- **Project management:** SCRUM methodologies, Agile.
- **Soft skills:** Leadership, Teamwork, Problem-solving, Communication, Mentoring.

Training and Certifications

- **CITI Program, Human Research, Social, Behavioral, and Educational Research (IRB Training), Basic Course.** Northern Arizona University, 2025.
- Java Programming Certification, IIT Bombay.
- C Programming Certification, IIT Bombay.

Extracurricular and Volunteer Experience

Volunteer at Game Jam *NAU Summer Workshop, 2023*

Participated in a summer workshop, Game Jam, organized by NAU for high school students to introduce them to game development and coding.

Responsibilities

- Guided and mentored high school students in developing their own games, from conceptualization to implementation.
- Facilitated workshops on basic game design principles, programming with game development tools, and collaborative project development.
- Assisted in organizing and managing event logistics, ensuring a smooth and productive workshop experience for all participants.

Impact

- Inspired and empowered young students to explore careers in technology and game development.
- Enhanced skills in mentorship, event coordination, and public speaking.

College Basketball Player *2015 – 2017*

Team Player and Competitor

- Played college-level basketball for two years, actively competing in regional tournaments and representing the college with dedication and sportsmanship.
- Demonstrated strong teamwork skills by collaborating with teammates to develop strategies and execute game plans effectively.

Discipline and Commitment

- Maintained a rigorous training schedule, balancing academic responsibilities with athletic commitments to excel in both areas.

- Showcased discipline and perseverance by consistently working on improving skills, fitness, and game strategies.

Achievement and Recognition

- Contributed to the team's success by playing key roles in crucial matches, earning recognition from coaches, teammates, and the college community.
- Participated in high-stakes tournaments, gaining valuable experience and showcasing resilience in competitive environments.

Student Coordinator, Vignana Jyothi Prajwal *Undergraduate*

Event Organization and Management

- Coordinated events for Vignana Jyothi Prajwal, including academic workshops, cultural programs, and technical seminars.
- Ensured smooth execution by meticulously planning and managing all logistical aspects, from venue selection to resource allocation.

Communication and Coordination

- Served as the primary point of contact between event organizers, participants, and external stakeholders, ensuring clear and effective communication.
- Handled inquiries and resolved issues during planning and execution, ensuring a seamless experience for all involved.

Impact and Personal Growth

- Successfully organized and executed multiple events that received positive feedback from participants and faculty, enhancing the reputation of Vignana Jyothi Prajwal.
- Developed strong organizational, leadership, and interpersonal skills through hands-on experience in event coordination and team management.
- Learned to thrive in high-pressure environments, balancing multiple tasks and priorities to deliver successful outcomes.